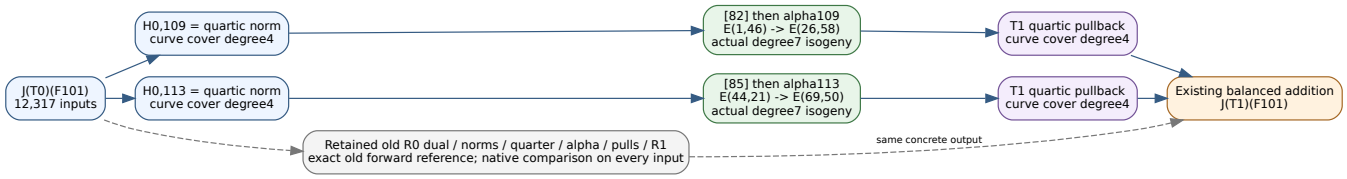
The first Jacobian seven-transfer using F101 arithmetic

2026-11-12 · fixed T0→T1 forward evaluator · scientific standing unsigned

The existing forward Psi now has a constructor and evaluator that use F101 arithmetic throughout. Native prototype15/PCC72e1d8 matches every 12,317 concrete output of the old transfer,with bijection,all fixed-partner sums,negations,identity and wrong-model controls. Selected alpha numerator, denominator,kernel and curve coefficients match the old maps exactly. This is an evaluator of the existing degree49 Jacobian map. [pcc:unverified]



Exact stages and their acceptance

The source and destination are the actual nonsplit sextics T0=[31,75,74,54,71,55,1] and T1=[95,26,19,27,94,91,1]. Both rational Jacobian groups have order12,317=109·113. The constructor takes accepted fixed covers and the selected elliptic kernel cubics[72,100,35,1] and [59,46,8,1]. It reuses existing Kohel and prime-field engines;parameter discovery remains inherited input provenance. [pcc:unverified]

Stage	Existing-engine evidence
Both T0 norms H0	Every12,317 old output/group/exception controls;585f35/2fcf0b
Quarter and alpha	Public inverse4 scalars82/85;coefficient-identical selected degree7 maps
T1 pullbacks K1	All109/113 inputs,old RiPi,norm4/off-diagonal0;5143f5
Balanced assembly	Original reduction arithmetic body byte-exact after extraction

All four T0/T1 pullbacks additionally pass49,300 ordered elliptic-group additions. Actual infinity multiplicities are[1,2] for T0 and[2,1] for T1. The two T1 covers each have116 finite-valued witnesses and two rational pole witnesses;all four geometric pole branches are retained. Separate flat-field replays5ab2bd/5eb5b1/850e92 validate the exact equations, degrees,complete source domains,infinities and poles. [pcc:unverified]

Why the replacement computes the same transfer

The accepted global factorization is4Psi=K1 alpha H0. Each H0 component is the pushforward of its exact degree4 cover;each K1 component is the corresponding T1 pullback. On the actual elliptic rational groups, 4·82=1 modulo109 and4·85=1 modulo113. Moving these public quarter scalars before alpha therefore gives Psi on J(T0)(F101),since4 is invertible on the order12,317 destination group. This uses no discrete-log labels and does not propose a global inverse morphism of[4]. The old adjoint norm7 preserves the original109/113 DLP components and recovery. Upstream PRIME_FIELD_PSI_PROOF.md/78176c records the complete argument. [pcc:unverified]

The curve maps are identified globally using exact polynomial equations, degree4 and a proved difference-degree bound16. More than16 distinct agreement points force equality. Old quotient/Picard/polarization/norm facts and expected-value provenance remain explicit dependencies. A negative ordinate passes the curve equation but fails the witnesses; equation fitting alone would not identify the map. [pcc:unverified]

Pullback jets and exact infinity bookkeeping

For finite P=(X,Y),form U=N-XQ^2,V=YQ^3/M moduloU. In the finite support algebra,F M^2=Y^{2}Q^6. Supplied units Yi,Qi give the explicit inverse M inverse=F M Yi^2 Qi^6,even if the quotient has nilpotents. Two bound QuarticFibre lemmas d4010b/4286c9 prove the actual cleared factorization and inverse,local kernel/axiom gate3413a4:2theorems,zero trusted native. That proof-stage census136written helpers/25modules remains partial formalization. [pcc:unverified]

Since divQ=f*(O)-2(infinity++infinity-),a generic degree4 fibre has balanced plus count-1. At P=+q,a degree dropk gives plus=k-1;at P=-q it gives plus=-1. The Y sign is essential. The implemented rational elliptic groups have odd order;geometric nonzero2-torsion needs another chart and is outside this implemented domain. QUARTIC_PULLBACK_PROOF.md/ 4934b4 supplies the divisor derivation and original-engine custody. [pcc:unverified]

Actual production integration and complete acceptance

The actual SevenChain.forward now selects the prime-field entry. Its exact old body remains forward_reference;the new complete comparison calls that explicit reference. Original dual,naive,source,geometric and chained controls remain. Complete changed check.sh04 exits0:25native/90reference unit+five integrations,one pre-existing ignored benchmark,all source guards, fmt and strictClippy. PCC76871d pins the whole131756-byte raw log as standardXZ14932bytes,raw/archive hashes/exact roundtrip and all eight unchanged post-run source02 hashes. [pcc:unverified]

The workspace restart interrupted03 before reference completion. Its full available partial48451bytes,metadata and sourcehash receipt remain frozen underf87a99;04 retries the identical source and complete check. No partial run is counted as passed. The new ready child targets immediate1505 at6cf; its own exact-headCI is separate. Parent1505 focused43165729371 succeeds 25native/85reference+five,fullfd53e1;parent1501 remains fixed5ceb/green. [pcc:unverified]

Written PRIME_FIELD_PSI.md,QUARTIC_PULLBACKS.md and T0_SECOND_QUARTIC.md copy upstream78176c/4934b4/a38aa4. Independent curve certificates and the faithful finite-algebra lemmas remain explicit dependencies;actual geometric formalization is partial. Existing reducer/old-chain bodies have exact extraction custody. The isolated upstream adjoint/T2/chain-compression results are outside this consumer checkpoint. [pcc:unverified]

Matched cold construction,transfer,solve,recovery and verification costs await S3/profile/livebudget admission. No timing gain is claimed. KB/ecdb unavailable;autoresearcher atlas vault locked. Frozen ledger forks and parent1484 C/conflict findings persist;integrate bottom-up after actual gates,retarget/rerun after parents land. [pcc:unverified]